

Telco churn analysis

From customer patterns to a measurable retention pilot

A transparent baseline. A focused next step.

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SQL

Python

Logistic regression

Business analysis



Which customers should a retention team review first?

Use historical patterns to prioritize outreach, then test whether the intervention works.

7,043

customer records

26.5%

labelled as churned

18

input features

Decision to support

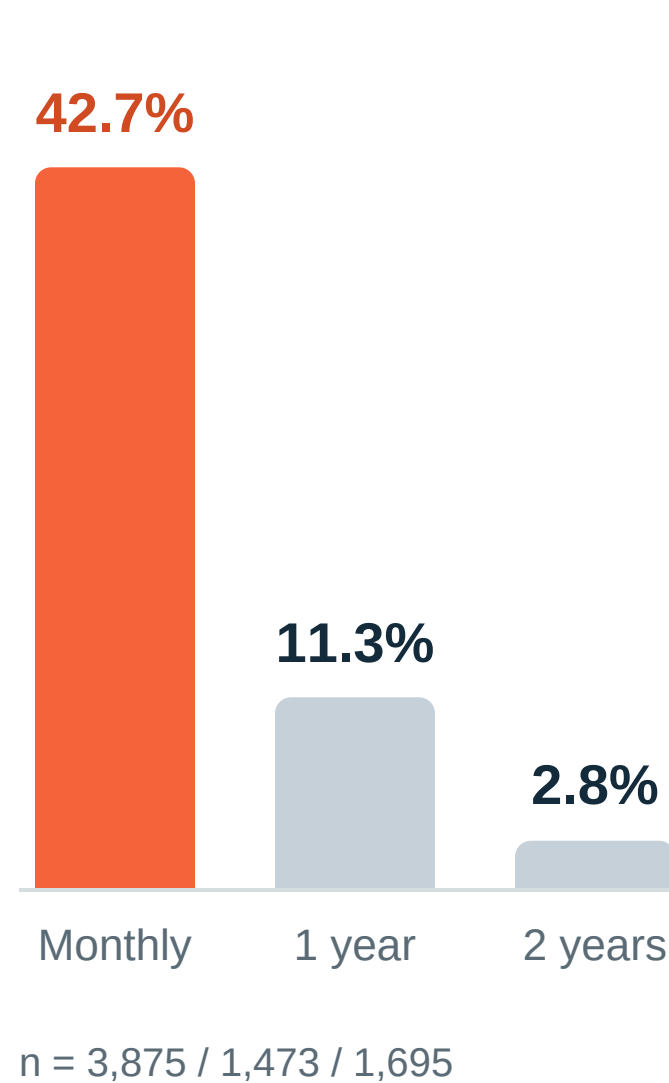
Allocate limited retention capacity using model scores and customer context.

Scope: a dataset snapshot; a future prediction horizon has not been validated.

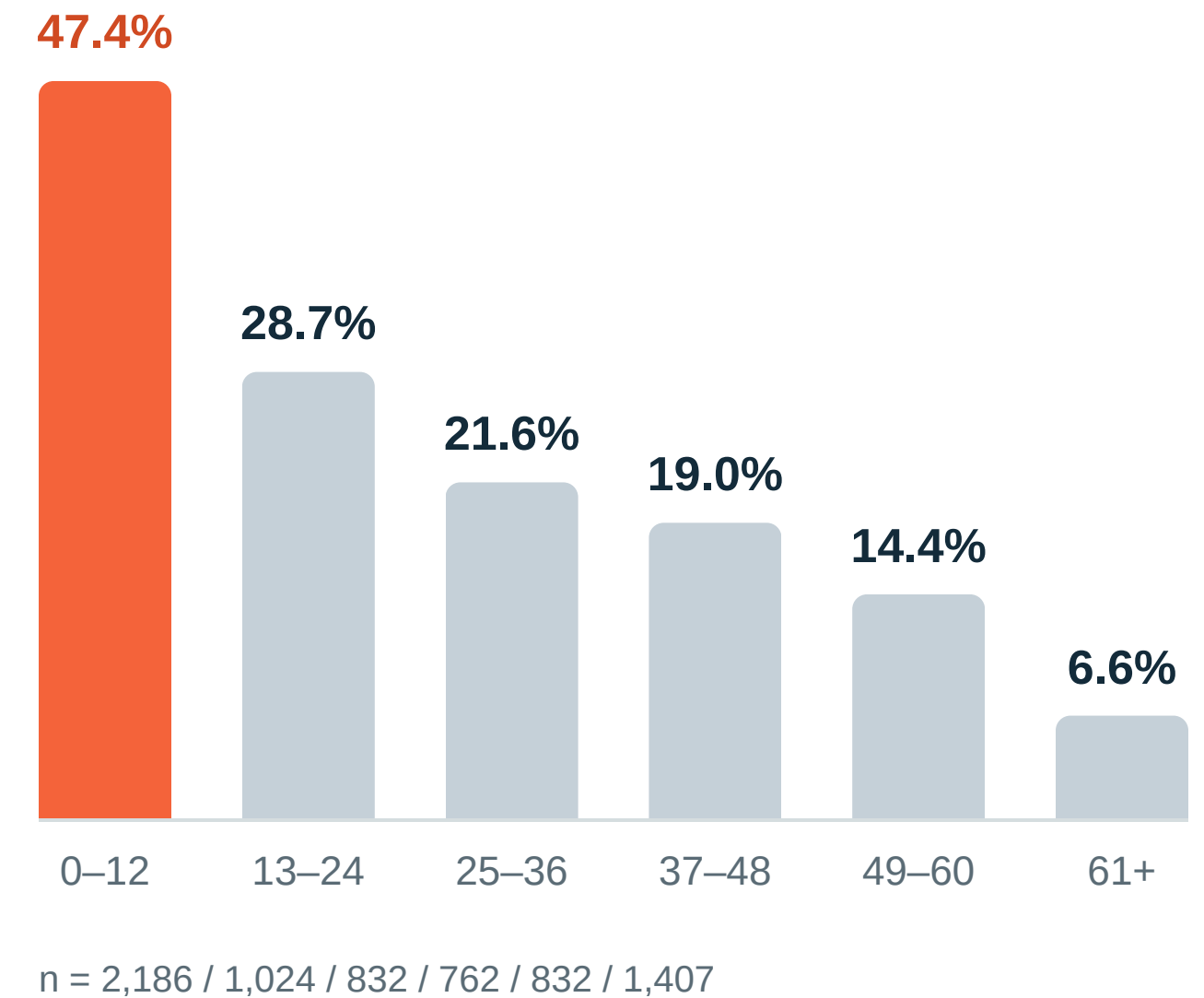
Churn is concentrated in short contracts and early tenure

Observed churn rates within each segment. These comparisons do not establish causality.

By contract



By tenure in months



An interpretable baseline with a separate test set

Logistic regression classifies the recorded churn label using customer and service attributes.

01 Prepare

18 inputs

Customer profile, tenure, services, contract and billing.

Exclude IDs, location and churn reasons.

02 Fit

5,634 training records

Scale numeric features; encode categories; fit logistic regression.

Preprocessing is fitted on training data only.

03 Evaluate

1,409 test records

Assess classification at 0.50 and ranking across all scores.

Random 80/20 split; seed 42. No time-based test yet.

Useful signal, with meaningful false positives and misses

Held-out test set: 1,409 customers, including 400 churners · classification threshold 0.50

68.1%

Precision



231 of 339 flagged customers are labelled as churned.

57.8%

Recall



231 of 400 churners are correctly flagged.

0.853

ROC AUC



Ranking quality across all score thresholds.

At this threshold: 108 false positives and 169 missed churners.

The highest-scoring 10% contain 2.75× the average churn share

A capacity-based view of the same test set; customers are ranked by unrounded model scores.



Implication: choose review volume around team capacity, then validate economics.

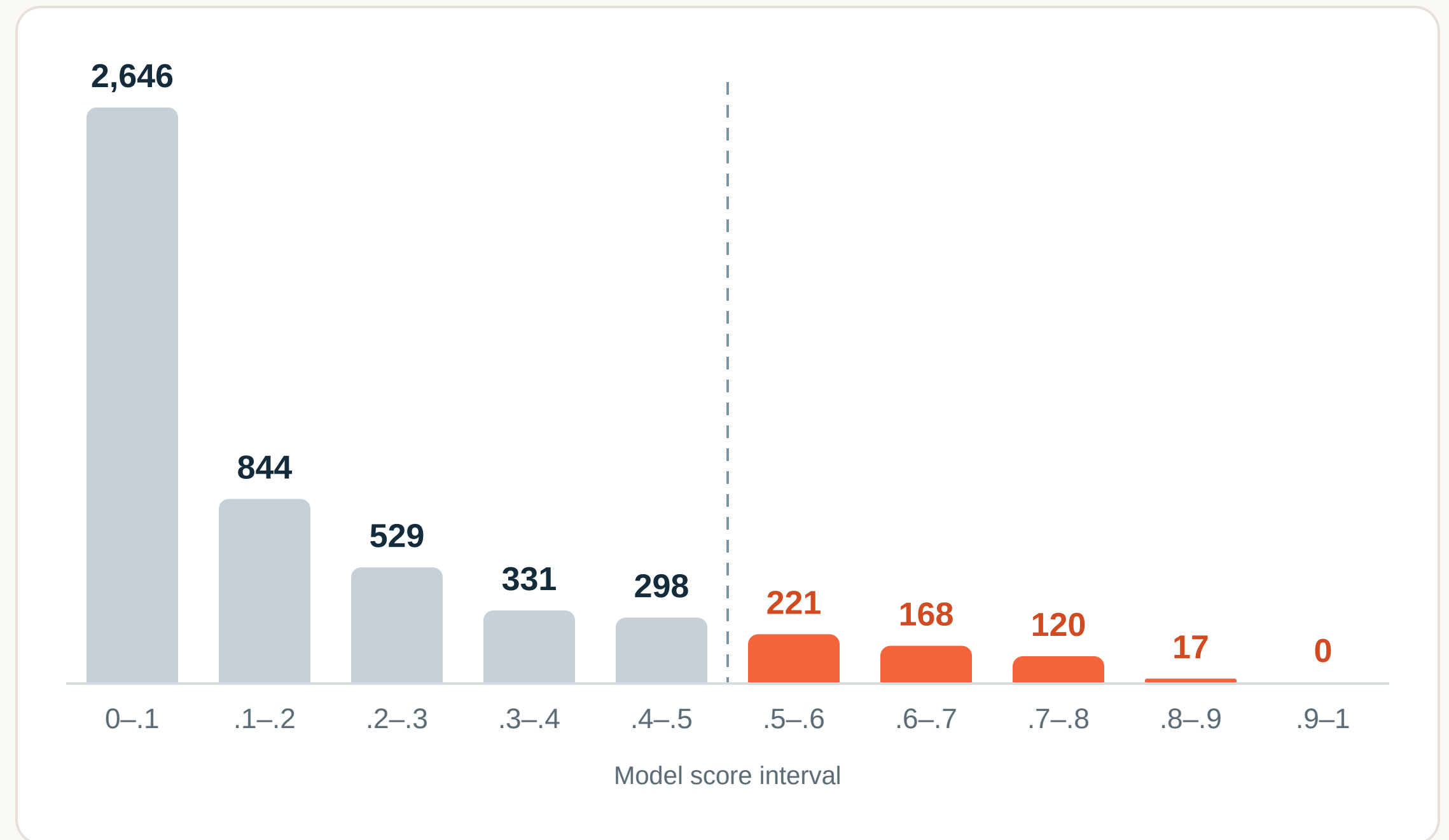
511 non-churned records receive scores above 0.50

Illustrative review volume: 9.9% of the 5,174 records labelled “No”.

Interpret with care

The scored file includes training records.

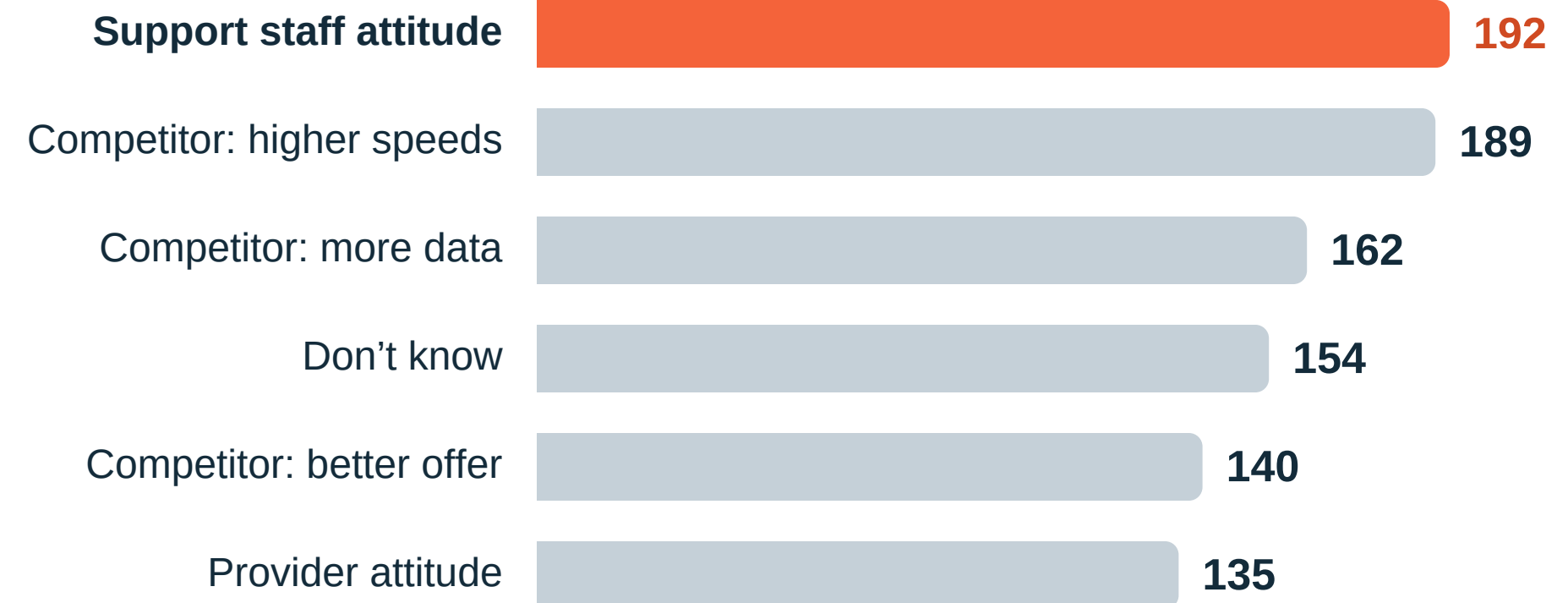
These are review candidates, not confirmed future churners.



Bins include their lower bound. The count of 511 uses scores strictly > 0.50 , rounded to 2 decimals.

Service interactions and competing offers deserve investigation

Six most frequent recorded reasons among 1,869 churned customers · counts, not causal effects



192

**cite support staff attitude
(10.3% of churners)**

Use these responses to shape interviews and pilot offers.

Translate the evidence into three focused retention tests

Prioritize eligible customers by score; test interventions against a randomized control group.

Evidence	Test	Primary measure
Support attitude is the most frequent reason	> Service recovery callback after a poor interaction	> Incremental retention net of support cost
Churn is highest in the first 12 months	> Proactive onboarding and technical assistance	> Retention lift and cost per retained customer
Monthly contracts and competing offers stand out	> Tailored plan review with a limited offer test	> Incremental margin after discount cost
Avoid blanket discounts. Set the offer and success criteria before running each test.		

Validate the decision before scaling retention outreach

A sensible next step is a small pilot with measurable retention and margin outcomes.

01

Define the prediction window

Confirm which inputs are available before churn can occur.

02

Validate future performance

Use a time-based holdout; assess calibration and segment stability.

03

Set contact capacity and economics

Compare score cutoffs with contact cost, margin and expected benefit.

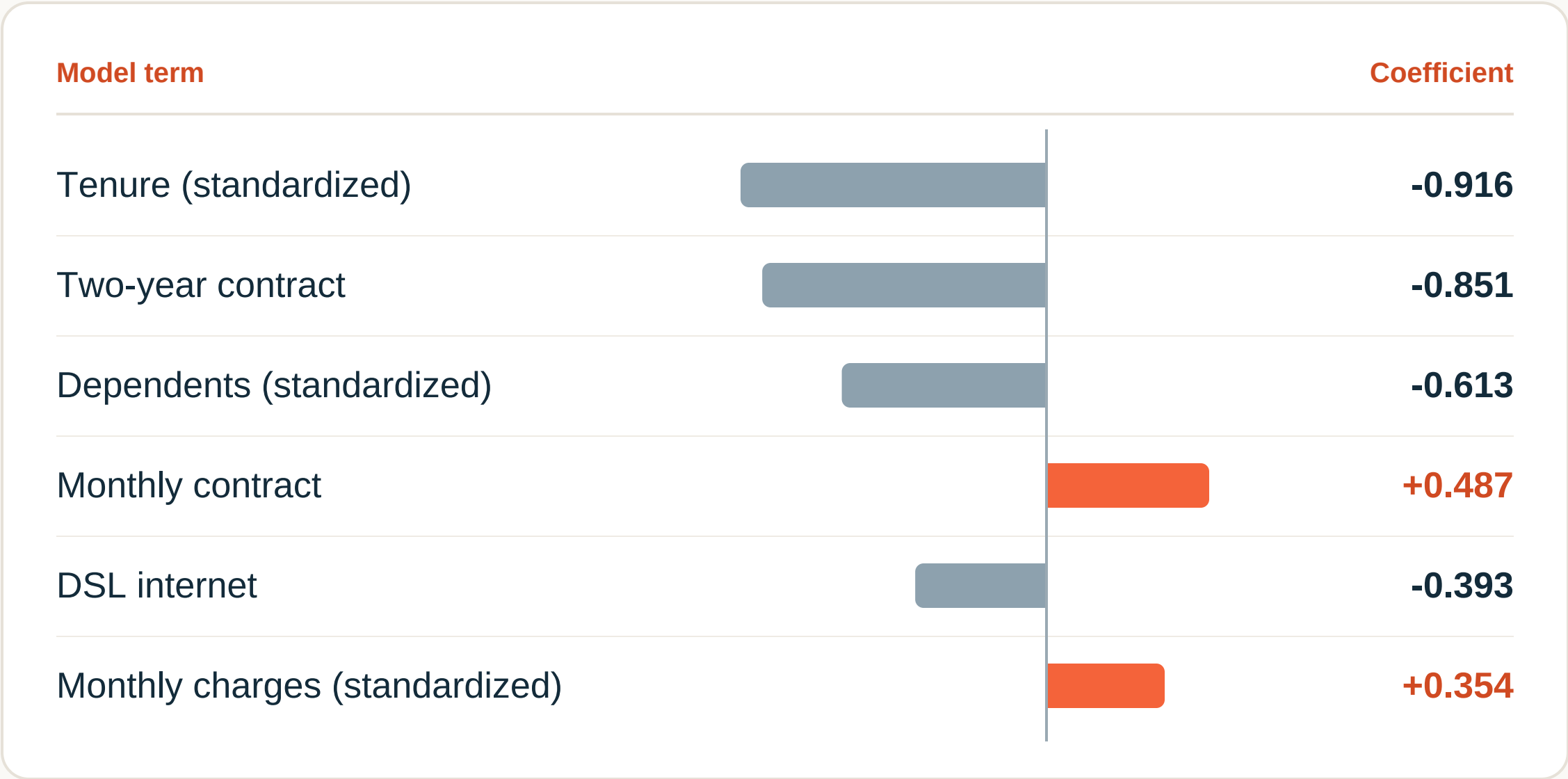
04

Measure incremental impact

Run a randomized pilot; scale only if the measured net value is positive.

Coefficient direction supports hypotheses, not causal claims

Selected fitted coefficients from the supplied model · negative means lower fitted log-odds



Read within this model

Scaling and one-hot encoding affect coefficient size.

Use associations to design follow-up analysis and tests.

What was evaluated — and what remains to be established

Reproduced baseline

- ✓ 80/20 random split, seed 42; no stratification.
- ✓ Logistic regression, L2, C = 1, max_iter = 1,000.
- ✓ Accuracy 80.3%; F1 0.625; average precision 0.682.
- ✓ Majority-class test baseline: 71.6% accuracy.

Still open before deployment

- Feature timing and a future prediction window.
- Time-based validation and calibrated scores.
- Thresholds tied to costs and contact capacity.
- Incremental impact of retention interventions.

Inputs

Customer profile, tenure, phone/internet services, service add-ons, contract and payment attributes.

Reproduction note

Recalculated using the original model and split.
Source code and data were not changed.